The Itô Formula: Intuition, Rules, and Applications

Lecture 4 Notes — Class Support and Student Summary

Goals and logical structure

Brownian quadratic variation turns the familiar chain rule into a second-order calculus. The lecture proceeds through one reusable sequence:

$$\text{quadratic variation} \longrightarrow \text{multiplication table} \longrightarrow \text{Itô formula} \longrightarrow \text{SDE transformations and finance.}$$

By the end, we can transform one- and two-factor Itô processes, solve GBM and OU models, compute the mean of a CIR process, and explain why correlation, convexity, discounting, and delta hedging all create or remove drift terms. Exercises are placed immediately after the relevant rule; complete short solutions follow them.

1 Quadratic Variation Becomes a Calculus

1.1 The correction already appeared in Lecture 3

Lecture 3 established the identity

$$\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T),$$

or equivalently

$$W_T^2 = 2\int_0^T W_t \, \mathrm{d}W_t + T.$$

The first term is the one predicted by the ordinary chain rule. The extra T is the quadratic variation $[W]_T$.

On one interval,

$$(W_t + \Delta W)^2 - W_t^2 = 2W_t\Delta W + (\Delta W)^2.$$

Brownian scaling gives $\Delta W = O_{\mathbb{P}}(\sqrt{\Delta t})$, so $(\Delta W)^2 = O_{\mathbb{P}}(\Delta t)$. The second-order terms are individually small, but there are order $1/\Delta t$ of them. Their accumulated effect is therefore order one.

One-step term	Typical size	Sum over n steps	Limit scale
Δt	n^{-1}	1	survives
$(\Delta W)^2$	n^{-1}	1	survives
$\Delta t\,\Delta W$	$n^{-3/2}$	$n^{-1/2}$	vanishes
$(\Delta t)^2$	n^{-2}	n^{-1}	vanishes

Bookkeeping principle

Retain every term whose sum can accumulate at order $\mathrm{d}t$. For a continuous diffusion this means keeping first-order terms and the quadratic products of Brownian increments.

1.2 Covariation and the differential tables

Definition 1.1 (Quadratic covariation)**.** For continuous processes X, Y and a partition $\pi = \{0 = t_0 < \cdots < t_n = t\}$, define

$$[X, Y]_t^{\pi} = \sum_{i=0}^{n-1} \big(X_{t_{i+1}} - X_{t_i}\big)\big(Y_{t_{i+1}} - Y_{t_i}\big).$$

When these sums converge in probability as the mesh tends to zero, their limit is denoted by $[X, Y]_t$. We write $[X]_t = [X, X]_t$.

The basic examples are

$$[W]_t = t, \qquad [W, t]_t = 0, \qquad [t]_t = 0.$$

The differential notation $\mathrm{d}X_t\,\mathrm{d}Y_t := \mathrm{d}[X, Y]_t$ summarizes these limiting identities. For one Brownian driver,

$\times$	$\mathrm{d}t$	$\mathrm{d}W_t$
$\mathrm{d}t$	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$

$$\iff \qquad (\mathrm{d}t)^2 = 0, \quad \mathrm{d}t\,\mathrm{d}W_t = 0, \quad (\mathrm{d}W_t)^2 = \mathrm{d}t.$$

The table is a mnemonic for limits

These are not literal algebraic infinitesimals. For example, $(\mathrm{d}W_t)^2 = \mathrm{d}t$ means that accumulated squared Brownian increments converge to elapsed time.

Let W, B be Brownian motions with constant correlation ρ:

$$[W, B]_t = \rho t, \qquad -1 \le \rho \le 1.$$

The two-factor table is

$\times$	$\mathrm{d}t$	$\mathrm{d}W_t$	$\mathrm{d}B_t$
$\mathrm{d}t$	0	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$	$\rho\,\mathrm{d}t$
$\mathrm{d}B_t$	0	$\rho\,\mathrm{d}t$	$\mathrm{d}t$

and, for a Brownian vector,

$$\mathrm{d}W_t^i\,\mathrm{d}W_t^j = \rho_{ij}\,\mathrm{d}t.$$

Example — correlation becomes drift in a product

The product identity gives

$$\mathrm{d}(W_t B_t) = W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t + \mathrm{d}W_t\,\mathrm{d}B_t = W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t + \rho\,\mathrm{d}t.$$

On a bounded horizon the stochastic integrals have mean zero, so

$$\mathbb{E}[W_t B_t] = \rho t,$$

which agrees with the covariance of the jointly Gaussian pair.

Exercise — a first table drill

Let

$$\mathrm{d}X_t = 2\,\mathrm{d}t + 3\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = -\,\mathrm{d}t + 4\,\mathrm{d}W_t, \qquad \mathrm{d}Z_t = -\,\mathrm{d}t + 4\,\mathrm{d}B_t.$$

Compute $(\mathrm{d}X_t)^2$, $\mathrm{d}X_t\,\mathrm{d}Y_t$, and $\mathrm{d}X_t\,\mathrm{d}Z_t$. Treat first the independent case and then the case $\mathrm{d}W_t\,\mathrm{d}B_t = -\frac{1}{2}\,\mathrm{d}t$.

Short solution

Every product containing $\mathrm{d}t$ vanishes. Therefore

$$(\mathrm{d}X_t)^2 = 9\,\mathrm{d}t, \qquad \mathrm{d}X_t\,\mathrm{d}Y_t = 12\,\mathrm{d}t, \qquad \mathrm{d}X_t\,\mathrm{d}Z_t = 12\rho\,\mathrm{d}t.$$

The last answer is 0 under independence and $-6\,\mathrm{d}t$ when $\rho = -1/2$. Only diffusion loadings and correlation enter covariation.

Exercise — general covariation formula

Suppose

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t + c_t\,\mathrm{d}B_t, \qquad \mathrm{d}Y_t = p_t\,\mathrm{d}t + q_t\,\mathrm{d}W_t + r_t\,\mathrm{d}B_t,$$

with $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$. Find $\mathrm{d}[X,Y]_t$.

Short solution

$$\boxed{\mathrm{d}[X,Y]_t = \big(b_t q_t + c_t r_t + \rho b_t r_t + \rho c_t q_t\big)\,\mathrm{d}t.}$$

The coefficients a_t, p_t do not appear because finite-variation terms have zero covariation with continuous semimartingales.

2 The Itô Formula Is a Second-Order Chain Rule

2.1 Heuristic and precise statement

For a smooth $f(t,x)$, Taylor expansion suggests

$$\Delta f \approx f_t \Delta t + f_x \Delta W + \frac{1}{2} f_{xx} (\Delta W)^2.$$

The mixed term $\Delta t \Delta W$ and the cubic term $(\Delta W)^3$ have accumulated order tending to zero. Replacing the quadratic term by Δt leads to the correct formula.

Theorem 2.1 (Brownian Itô formula)**.** Let $f \in C^{1,2}([0,T]\times\mathbb{R})$. Then

$$f(t,W_t) = f(0,W_0) + \int_0^t f_t(s,W_s)\,\mathrm{d}s + \int_0^t f_x(s,W_s)\,\mathrm{d}W_s + \frac{1}{2}\int_0^t f_{xx}(s,W_s)\,\mathrm{d}s.$$

Equivalently,

$$\boxed{\mathrm{d}f(t,W_t) = \left(f_t + \frac{1}{2} f_{xx}\right)(t,W_t)\,\mathrm{d}t + f_x(t,W_t)\,\mathrm{d}W_t.}$$

Construction idea. On a partition $0 = t_0 < \cdots < t_n = t$, telescope:

$$f(t,W_t) - f(0,W_0) = \sum_i \big(f(t_{i+1},W_{t_{i+1}}) - f(t_i,W_{t_i})\big).$$

A Taylor expansion at (t_i, W_{t_i}) gives

$$\Delta f_i = f_t \Delta t_i + f_x \Delta W_i + \frac{1}{2} f_{xx} (\Delta W_i)^2 + r_i.$$

The first sum converges to $\int f_t\,\mathrm{d}t$. The left-point second sum converges to the Itô integral $\int f_x\,\mathrm{d}W$. Quadratic variation turns the third sum into $\frac{1}{2}\int f_{xx}\,\mathrm{d}t$. After localization, smoothness controls the derivatives on compact sets and the accumulated remainder converges to zero in probability. □

Four-step calculation recipe

1. Treat t and x as deterministic variables.
2. Compute $f_t(t,x)$, $f_x(t,x)$, and $f_{xx}(t,x)$.
3. Substitute $x = W_t$.
4. Group the $\mathrm{d}t$ and $\mathrm{d}W_t$ terms separately.

Differentiate with respect to the placeholder x, then substitute the process. This simple notation discipline prevents many errors.

2.2 First examples and Brownian moments

Example — three immediate transforms

For $f(x) = x^2$,

$$\mathrm{d}(W_t^2) = 2W_t\,\mathrm{d}W_t + \mathrm{d}t.$$

Because

$$\mathbb{E}\int_0^T (2W_s)^2\,\mathrm{d}s = 4\int_0^T s\,\mathrm{d}s = 2T^2 < \infty,$$

the process $W_t^2 - t$ is a true martingale.
For $f(x) = x^3$,

$$\mathrm{d}(W_t^3) = 3W_t^2\,\mathrm{d}W_t + 3W_t\,\mathrm{d}t.$$

For $f(x) = \sin x$,

$$\mathrm{d}(\sin W_t) = \cos(W_t)\,\mathrm{d}W_t - \frac{1}{2}\sin(W_t)\,\mathrm{d}t.$$

Even though W has zero drift, curvature can create drift in $f(W)$.

Exercise — manufacture a fourth-order martingale

Apply the formula to W_t^4. Find c such that

$$W_t^4 - c\int_0^t W_s^2\,\mathrm{d}s$$

has zero drift, and recover $\mathbb{E}[W_t^4]$.

Short solution

$$\mathrm{d}(W_t^4) = 4W_t^3\,\mathrm{d}W_t + 6W_t^2\,\mathrm{d}t.$$

Thus

$$M_t = W_t^4 - 6\int_0^t W_s^2\,\mathrm{d}s$$

has zero drift. Since $\mathbb{E}\int_0^T 16W_s^6\,\mathrm{d}s < \infty$, it is a true martingale on $[0,T]$. Hence

$$\mathbb{E}[W_t^4] = 6\int_0^t \mathbb{E}[W_s^2]\,\mathrm{d}s = 6\int_0^t s\,\mathrm{d}s = 3t^2.$$

Exercise — one more step gives the sixth moment

Use $\mathbb{E}[W_t^4] = 3t^2$ and the Itô formula for x^6 to compute $\mathbb{E}[W_T^6]$.

Short solution

$$\mathrm{d}(W_t^6) = 6W_t^5\,\mathrm{d}W_t + 15W_t^4\,\mathrm{d}t.$$

The integrand is square-integrable on bounded horizons because Brownian motion has finite moments of every order. Taking expectations,

$$\boxed{\mathbb{E}[W_T^6] = 15\int_0^T \mathbb{E}[W_t^4]\,\mathrm{d}t = 45\int_0^T t^2\,\mathrm{d}t = 15T^3.}$$

More generally, $m_n(t) = \mathbb{E}[W_t^n]$ satisfies

$$m_n'(t) = \frac{n(n-1)}{2}m_{n-2}(t).$$

2.3 Stopping turns martingales into exit-time identities

For $a > 0$, define the first exit time

$$\tau_a = \inf\{t \geq 0 : |W_t| = a\}.$$

The safe order of operations is to stop first at the bounded time $\tau_t = t \wedge \tau_a$, apply optional sampling, and only then let $t \to \infty$.

Exercise — exit time and occupation inside an interval

Use

$$W_t^2 - t \qquad \text{and} \qquad W_t^4 - 6\int_0^t W_s^2\,\mathrm{d}s$$

to determine

$$\mathbb{E}[\tau_a], \qquad \mathbb{E}\left[\int_0^{\tau_a} W_s^2\,\mathrm{d}s\right].$$

Short solution

Optional sampling at τ_t gives

$$\mathbb{E}[W_{\tau_t}^2] = \mathbb{E}[\tau_t] \leq a^2.$$

Monotone convergence yields $\mathbb{E}[\tau_a] \leq a^2$, hence $\tau_a < \infty$ almost surely. Brownian continuity gives $|W_{\tau_a}| = a$. Since $|W_{\tau_t}| \leq a$, dominated convergence then yields

$$\boxed{\mathbb{E}[\tau_a] = a^2.}$$

Stopping the fourth-order martingale gives

$$\mathbb{E}[W_{\tau_t}^4] = 6\mathbb{E}\left[\int_0^{\tau_t} W_s^2\,\mathrm{d}s\right].$$

Dominated convergence on the left and monotone convergence on the right give

$$\boxed{\mathbb{E}\left[\int_0^{\tau_a} W_s^2\,\mathrm{d}s\right] = \frac{a^4}{6}.}$$

2.4 Exponentials and the backward heat equation

Proposition 2.2 (Exponential martingale)**.** For $\lambda \in \mathbb{R}$,

$$M_t = \exp\left(\lambda W_t - \frac{1}{2}\lambda^2 t\right)$$

satisfies

$$\mathrm{d}M_t = \lambda M_t\,\mathrm{d}W_t.$$

Moreover $M_t \geq 0$ and a direct Gaussian calculation gives $\mathbb{E}[M_t] = 1$, so M is a true martingale.

Proof. For $f(t,x) = e^{\lambda x - \lambda^2 t/2}$,

$$f_t = -\frac{1}{2}\lambda^2 f, \qquad f_x = \lambda f, \qquad f_{xx} = \lambda^2 f.$$

The two drift terms cancel in the Itô formula. Also,

$$\mathbb{E}[M_t] = e^{-\lambda^2 t/2}\mathbb{E}[e^{\lambda W_t}] = e^{-\lambda^2 t/2} e^{\lambda^2 t/2} = 1.$$

□

More generally, for $Y_t = u(t, W_t)$,

$$\mathrm{d}Y_t = \left(u_t + \frac{1}{2}u_{xx}\right)(t, W_t)\,\mathrm{d}t + u_x(t, W_t)\,\mathrm{d}W_t.$$

Thus the drift vanishes when

$$\boxed{u_t + \frac{1}{2}u_{xx} = 0.}$$

This is the backward heat equation. Suitable growth or integrability conditions are still needed to upgrade the resulting local martingale to a true martingale.

Exercise — which transforms have zero drift?

For fixed $T > 0$, test

$$u_1(t,x) = x^2 - t, \quad u_2(t,x) = x^2 + t, \quad u_3(t,x) = x^2 + T - t, \quad u_4(t,x) = e^{\lambda x - \lambda^2 t/2}.$$

Compute $u_t + \frac{1}{2}u_{xx}$ and interpret $u_3(t, W_t)$.

Short solution

u	$u_t + \frac{1}{2}u_{xx}$	conclusion
$x^2 - t$	0	martingale
$x^2 + t$	2	drift $2\,dt$
$x^2 + T - t$	0	martingale
$e^{\lambda x - \lambda^2 t/2}$	0	martingale

Since the third process equals W_T^2 at $t = T$,

$$W_t^2 + T - t = \mathbb{E}[W_T^2 \mid \mathcal{F}_t].$$

Exercise — a conditional fourth moment

Find a polynomial $u(t,x)$ such that

$$u(T,x) = x^4, \qquad u_t + \frac{1}{2}u_{xx} = 0.$$

Deduce a formula for $\mathbb{E}[W_T^4 \mid \mathcal{F}_t]$.

Short solution

Let $h = T - t$. Direct differentiation shows that

$$u(t,x) = x^4 + 6hx^2 + 3h^2$$

solves the backward heat equation and has terminal value x^4. Therefore

$$\boxed{\mathbb{E}[W_T^4 \mid \mathcal{F}_t] = W_t^4 + 6(T-t)W_t^2 + 3(T-t)^2.}$$

This also follows by expanding $W_T = W_t + (W_T - W_t)$ and using Gaussian moments.

3 From Brownian Motion to Itô Processes

3.1 Definition, quadratic variation, and the general formula

Definition 3.1 (Itô process)**.** On a fixed horizon $[0,T]$, an Itô process has the representation

$$X_t = X_0 + \int_0^t a_s\,ds + \int_0^t b_s\,dW_s,$$

where, for example, a, b are progressively measurable and

$$\int_0^T |a_s|\,ds < \infty \quad \text{a.s.}, \qquad \mathbb{E}\left[\int_0^T b_s^2\,ds\right] < \infty.$$

We use the shorthand

$$dX_t = a_t\,dt + b_t\,dW_t.$$

The drift $a_t\,\mathrm{d}t$ is the local trend; the diffusion $b_t\,\mathrm{d}W_t$ is the local random shock. Applying the multiplication table,

$$(\mathrm{d}X_t)^2 = a_t^2(\mathrm{d}t)^2 + 2a_tb_t\,\mathrm{d}t\,\mathrm{d}W_t + b_t^2(\mathrm{d}W_t)^2 = b_t^2\,\mathrm{d}t.$$

Hence

$$\boxed{\mathrm{d}[X]_t = b_t^2\,\mathrm{d}t, \qquad [X]_t = \int_0^t b_s^2\,\mathrm{d}s.}$$

Quadratic variation records volatility, not drift.

Theorem 3.2 (One-dimensional Itô formula)**.** Let

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t, \qquad f \in C^{1,2}([0,T]\times\mathbb{R}).$$

Then

$$\boxed{\mathrm{d}f(t,X_t) = \left(f_t + a_tf_x + \frac{1}{2}b_t^2f_{xx}\right)(t,X_t)\,\mathrm{d}t + b_tf_x(t,X_t)\,\mathrm{d}W_t.}$$

Equivalently,

$$\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \frac{1}{2}f_{xx}\,\mathrm{d}[X].$$

Example — the square of a general diffusion

For $f(x) = x^2$,

$$\boxed{\mathrm{d}(X_t^2) = (2X_ta_t + b_t^2)\,\mathrm{d}t + 2X_tb_t\,\mathrm{d}W_t.}$$

Whenever the stochastic integral is a true martingale,

$$X_t^2 - X_0^2 - \int_0^t (2X_sa_s + b_s^2)\,\mathrm{d}s$$

is a martingale.

Exercise — arithmetic diffusion and its second moment

Let

$$X_t = x + \mu t + \sigma W_t.$$

Compute $\mathrm{d}(X_t^2)$, find a compensated process with zero drift, and calculate $\mathbb{E}[X_t^2]$.

Short solution

$$\mathrm{d}(X_t^2) = (2\mu X_t + \sigma^2)\,\mathrm{d}t + 2\sigma X_t\,\mathrm{d}W_t.$$

The stochastic integrand is square-integrable on bounded horizons because $\int_0^T \mathbb{E}[X_s^2]\,\mathrm{d}s < \infty$. Taking expectations and either solving the resulting integral equation or using the known Gaussian law gives

$$\boxed{\mathbb{E}[X_t^2] = (x+\mu t)^2 + \sigma^2 t.}$$

This is $(\text{mean})^2 + \text{variance}$.

When may a stochastic integral be erased in expectation?

If

$$\mathbb{E}\left[\int_0^T H_s^2\,\mathrm{d}s\right] < \infty,$$

then $\int_0^t H_s\,\mathrm{d}W_s$ is an L^2-martingale and has expectation zero. Pathwise local square-integrability normally gives only a local martingale. Zero drift alone is not enough. Absolute integrability is also needed before exchanging expectation with a drift integral.

3.2 Logarithms and inverses

Proposition 3.3 (Reusable log rule)**.** Suppose $X_t > 0$ and

$$\frac{\mathrm{d}X_t}{X_t} = \alpha_t\,\mathrm{d}t + \beta_t\,\mathrm{d}W_t.$$

Then

$$\boxed{\mathrm{d}\log X_t = \left(\alpha_t - \frac{1}{2}\beta_t^2\right)\,\mathrm{d}t + \beta_t\,\mathrm{d}W_t.}$$

Proof. For $f(x) = \log x$,

$$f'(x) = \frac{1}{x}, \qquad f''(x) = -\frac{1}{x^2}.$$

Substitution into the general formula gives the claim. The term $-\beta_t^2/2$ is often called volatility drag. □

If strict positivity is not known globally, the identity is valid only up to the first hitting time of zero.

Exercise — invert an Itô process

Let $X_t \neq 0$ and

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t.$$

Find $\mathrm{d}(1/X_t)$, then specialize to $\mathrm{d}X_t/X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$.

Short solution

With $f(x) = x^{-1}$,

$$f'(x) = -x^{-2}, \qquad f''(x) = 2x^{-3}.$$

Therefore

$$\boxed{\mathrm{d}\left(\frac{1}{X_t}\right) = \left(-\frac{a_t}{X_t^2} + \frac{b_t^2}{X_t^3}\right)\,\mathrm{d}t - \frac{b_t}{X_t^2}\,\mathrm{d}W_t.}$$

For geometric dynamics and $R_t = 1/X_t$,

$$\boxed{\frac{\mathrm{d}R_t}{R_t} = (\sigma^2 - \mu)\,\mathrm{d}t - \sigma\,\mathrm{d}W_t.}$$

The $+\sigma^2$ term is the correction that an ordinary quotient rule would miss. The product rule applied to $X_t R_t = 1$ verifies the answer.

4 Transformations Solve Canonical SDEs

4.1 Geometric Brownian motion

Consider

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0, \quad \sigma > 0.$$

The logarithm converts relative shocks into additive shocks:

$$\mathrm{d}\log S_t = \left(\mu - \frac{1}{2}\sigma^2\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Integration yields the explicit solution

$$\boxed{S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}.}$$

It is strictly positive, which justifies the logarithmic transformation. Moreover,

$$\log S_t \sim N\left(\log S_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t,\, \sigma^2 t\right),$$

so S_t is lognormal.

Proposition 4.1 (Mean, median, and a target probability)**.**

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \mathrm{Med}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

For $K > 0$,

$$\mathbb{P}(S_T > K) = 1 - \Phi\left(\frac{\log(K/S_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

Example — one year of GBM in numbers

Let

$$S_0 = 100, \qquad \mu = 8\%, \qquad \sigma = 20\%, \qquad T = 1.$$

Then

$$\mathbb{E}[S_1] = 100e^{0.08} \approx 108.33, \qquad \mathrm{Med}(S_1) = 100e^{0.06} \approx 106.18.$$

For the target $K = 110$,

$$\mathbb{P}(S_1 > 110) = 1 - \Phi\left(\frac{\log(1.10) - 0.06}{0.20}\right) \approx 43.0\%.$$

The mean, median, and target probability answer different questions. A positive expected return does not eliminate downside uncertainty.

Exercise — powers of a GBM

For $p \in \mathbb{R}$, let $Y_t = S_t^p$.

1. Find $\mathrm{d}Y_t/Y_t$.
2. Choose k such that $e^{-kt}S_t^p$ has zero drift.
3. Check $p = 0, 1, 2, -1$.

Short solution

$$\boxed{\frac{\mathrm{d}(S_t^p)}{S_t^p} = \left[p\mu + \frac{1}{2}p(p-1)\sigma^2\right]\mathrm{d}t + p\sigma\,\mathrm{d}W_t.}$$

Thus

$$k = p\mu + \frac{1}{2}p(p-1)\sigma^2.$$

For constant coefficients, $e^{-kt}S_t^p$ is a constant multiple of an exponential Brownian martingale and is therefore a true martingale. For $p = -1$, the drift is $\sigma^2 - \mu$, matching the inverse calculation.

If coefficients vary through time,

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t,$$

the same log transform gives

$$\boxed{S_t = S_0 \exp\left\{\int_0^t \left(\mu_s - \frac{1}{2}\sigma_s^2\right)\mathrm{d}s + \int_0^t \sigma_s\,\mathrm{d}W_s\right\}.}$$

This holds under the usual progressive measurability and local integrability conditions. If μ, σ are deterministic, the log price remains Gaussian.

4.2 Ornstein–Uhlenbeck and Vasicek mean reversion

The OU SDE is

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0.$$

Above m, the drift is negative; below m, it is positive. Multiplying by $e^{\kappa t}$ removes the linear drift. Because this integrating factor has finite variation,

$$\begin{aligned} \mathrm{d}(e^{\kappa t} X_t) &= e^{\kappa t}\,\mathrm{d}X_t + \kappa e^{\kappa t} X_t\,\mathrm{d}t \\ &= \kappa m e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t. \end{aligned}$$

Integrating between s and t gives

$$\boxed{X_t = m + (X_s - m)e^{-\kappa(t-s)} + \sigma \int_s^t e^{-\kappa(t-u)}\,\mathrm{d}W_u.}$$

Conditionally on $\mathcal{F}_s$, the stochastic integral is centered Gaussian. Therefore

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = m + (X_s - m)e^{-\kappa(t-s)}$$

and

$$\mathrm{Var}(X_t \mid \mathcal{F}_s) = \sigma^2 \int_s^t e^{-2\kappa(t-u)}\,\mathrm{d}u = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa(t-s)}\right).$$

Exercise — read the long-run behavior from the OU formula

Find the limiting conditional mean, limiting conditional variance, and half-life of the deviation from m. Evaluate the half-life at $\kappa = 0.7$ per year.

Short solution

$$\mathbb{E}[X_t \mid \mathcal{F}_s] \longrightarrow m, \qquad \mathrm{Var}(X_t \mid \mathcal{F}_s) \longrightarrow \frac{\sigma^2}{2\kappa}.$$

The half-life h solves $e^{-\kappa h} = 1/2$, hence

$$\boxed{h = \frac{\log 2}{\kappa}.}$$

When $\kappa = 0.7$, $h \approx 0.99$ year. After two years,

$$e^{-0.7\times 2} \approx 24.7\%$$

of the current deviation remains in the conditional mean.

4.3 The first moment of a CIR process

Assume a nonnegative process satisfies

$$\mathrm{d}R_t = (\alpha - \beta R_t)\,\mathrm{d}t + \sigma\sqrt{R_t}\,\mathrm{d}W_t, \qquad R_0 = r_0, \quad \alpha, \beta, \sigma > 0.$$

The integrating factor $e^{\beta t}$ gives

$$\mathrm{d}(e^{\beta t} R_t) = \alpha e^{\beta t}\,\mathrm{d}t + \sigma e^{\beta t}\sqrt{R_t}\,\mathrm{d}W_t.$$

After localization and the usual integrability check, the stochastic integral has mean zero. Hence

$$\mathbb{E}[e^{\beta t} R_t] = r_0 + \frac{\alpha}{\beta}(e^{\beta t} - 1),$$

so

$$\boxed{\mathbb{E}[R_t] = r_0 e^{-\beta t} + \frac{\alpha}{\beta}(1 - e^{-\beta t}) \longrightarrow \frac{\alpha}{\beta}.}$$

The square-root volatility controls dispersion, but it does not enter the first-moment ODE.

Exercise — optional extension: the CIR second moment

Apply the Itô formula to R_t^2. Obtain an ODE for $m_2(t) = \mathbb{E}[R_t^2]$ in terms of $m_1(t) = \mathbb{E}[R_t]$, and find the limiting variance.

Short solution

$$\mathrm{d}(R_t^2) = \left((2\alpha+\sigma^2)R_t - 2\beta R_t^2\right)\,\mathrm{d}t + 2\sigma R_t^{3/2}\,\mathrm{d}W_t.$$

Thus

$$m_2'(t) + 2\beta m_2(t) = (2\alpha+\sigma^2)m_1(t).$$

Using the integrating factor $e^{2\beta t}$,

$$\begin{aligned} m_2(t) = {} & r_0^2 e^{-2\beta t} \\ & + (2\alpha+\sigma^2)\left[\frac{\alpha}{2\beta^2}(1-e^{-2\beta t}) + \frac{r_0-\alpha/\beta}{\beta}(e^{-\beta t}-e^{-2\beta t})\right]. \end{aligned}$$

Therefore

$$\lim_{t\to\infty}\operatorname{Var}(R_t) = \frac{\alpha\sigma^2}{2\beta^2}.$$

5 Products, Ratios, and Several Risk Factors

5.1 The multidimensional formula and product rule

Theorem 5.1 (Two-dimensional Itô formula)**.** Let $f \in C^{1,2}([0,T]\times\mathbb{R}^2)$ and let X, Y be continuous Itô processes. Then

$$\begin{aligned} \mathrm{d}f(t,X_t,Y_t) = {} & f_t\,\mathrm{d}t + f_x\,\mathrm{d}X_t + f_y\,\mathrm{d}Y_t \\ & + \frac{1}{2}f_{xx}\,\mathrm{d}[X]_t + f_{xy}\,\mathrm{d}[X,Y]_t + \frac{1}{2}f_{yy}\,\mathrm{d}[Y]_t. \end{aligned}$$

The mixed derivative has no factor $1/2$: it appears twice in the symmetric Hessian sum. If the Brownian drivers have correlation ρ, the mixed term carries that correlation.

Choosing $f(x,y) = xy$ gives the stochastic product rule

$$\boxed{\mathrm{d}(X_tY_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}[X,Y]_t.}$$

For

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = c_t\,\mathrm{d}t + d_t\,\mathrm{d}B_t,$$

we have

$$\mathrm{d}[X,Y]_t = \rho b_t d_t\,\mathrm{d}t.$$

Example — product of two correlated assets

Suppose

$$\frac{\mathrm{d}S_t}{S_t} = \mu_S\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}V_t}{V_t} = \mu_V\,\mathrm{d}t + \sigma_V\,\mathrm{d}B_t,$$

and $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$. Then

$$\boxed{\frac{\mathrm{d}(S_tV_t)}{S_tV_t} = (\mu_S+\mu_V+\rho\sigma_S\sigma_V)\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t + \sigma_V\,\mathrm{d}B_t.}$$

Positive correlation increases the product drift through covariation; negative correlation decreases it.

Exercise — normalize a sum of Brownian motions

Let W, B be jointly Brownian with correlation $\rho > -1$. Find a constant c such that $c(W+B)$ is a standard Brownian motion. What happens at $\rho = -1$?

Short solution

$$\mathrm{d}[W+B]_t = (\,\mathrm{d}W_t + \mathrm{d}B_t)^2 = 2(1+\rho)\,\mathrm{d}t.$$

Hence

$$\boxed{\widetilde{W}_t = \frac{W_t + B_t}{\sqrt{2(1+\rho)}}}$$

has quadratic variation t. Joint Gaussianity, or Levy's characterization, shows that it is Brownian. At $\rho = -1$, $B = -W$ up to indistinguishability and the sum is identically zero.

5.2 Ratios and inverse corrections

Exercise — ratio of two risky assets

With the asset dynamics above, let $R_t = S_t/V_t$. Find $\mathrm{d}R_t/R_t$ using $R = S(1/V)$.

Short solution

First apply the inverse rule:

$$\frac{\mathrm{d}(1/V_t)}{1/V_t} = (\sigma_V^2 - \mu_V)\,\mathrm{d}t - \sigma_V\,\mathrm{d}B_t.$$

Then use the product rule:

$$\begin{aligned}\frac{\mathrm{d}R_t}{R_t} &= \frac{\mathrm{d}S_t}{S_t} + \frac{\mathrm{d}(1/V_t)}{1/V_t} + \frac{\mathrm{d}S_t}{S_t}\frac{\mathrm{d}(1/V_t)}{1/V_t} \\ &= \left(\mu_S - \mu_V + \sigma_V^2 - \rho\sigma_S\sigma_V\right)\,\mathrm{d}t \\ &\quad + \sigma_S\,\mathrm{d}W_t - \sigma_V\,\mathrm{d}B_t.\end{aligned}$$

Thus

$$\boxed{\frac{\mathrm{d}R_t}{R_t} = \left(\mu_S - \mu_V + \sigma_V^2 - \rho\sigma_S\sigma_V\right)\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t - \sigma_V\,\mathrm{d}B_t.}$$

If $S = V$, every term vanishes. If V is deterministic, $\sigma_V = 0$, ordinary quotient calculus returns.

Exercise — log ratio as a consistency check

Compute $\mathrm{d}\log(S_t/V_t)$ directly from the two log-price dynamics. Why does ρ not appear in its drift, even though it appears in the level-ratio drift?

Short solution

$$\boxed{\mathrm{d}\log\frac{S_t}{V_t} = \left[\mu_S - \mu_V - \frac{1}{2}(\sigma_S^2 - \sigma_V^2)\right]\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t - \sigma_V\,\mathrm{d}B_t.}$$

Correlation affects the quadratic variation of the final diffusion:

$$\mathrm{d}\left[\log(S/V)\right]_t = \left(\sigma_S^2 + \sigma_V^2 - 2\rho\sigma_S\sigma_V\right)\,\mathrm{d}t.$$

Applying the exponential rule to the log ratio recreates the $-\rho\sigma_S\sigma_V$ term in the level-ratio drift.

5.3 Minimum-variance hedging

Consider the instantaneous shock

$$\sigma_X\,\mathrm{d}W_t - h\sigma_Y\,\mathrm{d}B_t, \qquad \sigma_Y > 0, \qquad \mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t.$$

Its variance rate is

$$q(h) = \sigma_X^2 + h^2\sigma_Y^2 - 2h\rho\sigma_X\sigma_Y.$$

Differentiating,

$$q'(h) = 2h\sigma_Y^2 - 2\rho\sigma_X\sigma_Y.$$

Since $q''(h) = 2\sigma_Y^2 > 0$, the unique minimizer is

$$\boxed{h^* = \rho\frac{\sigma_X}{\sigma_Y}, \qquad q(h^*) = \sigma_X^2(1-\rho^2).}$$

Only the correlated component can be hedged; orthogonal noise remains.

Example — numerical hedge ratio

If

$$\sigma_X = 25\%, \qquad \sigma_Y = 20\%, \qquad \rho = 0.8,$$

then

$$h^* = 0.8\frac{0.25}{0.20} = 1, \qquad \sqrt{q(h^*)} = 25\%\sqrt{1-0.8^2} = 15\%.$$

The hedge removes the correlated component in this simplified instantaneous calculation, but it cannot remove the 15% orthogonal component.

5.4 Optional extension: predictable rotations

Let W be a standard d-dimensional Brownian motion and H_t a predictable matrix satisfying

$$H_t H_t^\top = I_d.$$

Define

$$B_t = \int_0^t H_s \, \mathrm{d}W_s.$$

Then B is a continuous local martingale and

$$\mathrm{d}[B^i, B^j]_t = (H_t H_t^\top)_{ij} \, \mathrm{d}t = \delta_{ij} \, \mathrm{d}t.$$

Levy's characterization implies that B is again a standard d-dimensional Brownian motion. In one dimension,

$$\int_0^t \mathrm{sgn}(W_s) \, \mathrm{d}W_s$$

is Brownian even though its integrand depends on the observed path.

6 Finance Applications of Itô Calculus

6.1 Discounting and the probability measure

Let the bank account and risky asset satisfy

$$\mathrm{d}A_t = rA_t \, \mathrm{d}t, \qquad \mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t^{\mathbb{P}}.$$

Since $A_t^{-1} = A_0^{-1} e^{-rt}$ has finite variation,

$$\mathrm{d}(A_t^{-1}) = -rA_t^{-1} \, \mathrm{d}t$$

and $\mathrm{d}[A^{-1}, S]_t = 0$. For $\widetilde{S}_t = S_t/A_t$,

$$\boxed{\frac{\mathrm{d}\widetilde{S}_t}{\widetilde{S}_t} = (\mu - r) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{P}}.}$$

Discounting by the bank account does not generally remove the physical drift.
Under a pricing measure $\mathbb{Q}$, the model is written

$$\frac{\mathrm{d}S_t}{S_t} = r \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{Q}},$$

so

$$\frac{\mathrm{d}\widetilde{S}_t}{\widetilde{S}_t} = \sigma \, \mathrm{d}W_t^{\mathbb{Q}}.$$

The discounted price is then a local martingale; standard integrability conditions make it a true martingale.

Always state the measure

The same asset can have drift μ under the physical measure and drift r under a pricing measure. The martingale property of discounted prices is a risk-neutral statement, not an automatic physical-measure identity.

6.2 Option dynamics and delta hedging

Suppose $V_t = v(t, S_t)$, with

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t.$$

The Itô formula gives

$$\boxed{\mathrm{d}V_t = \left(v_t + \mu S_t v_s + \frac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\mathrm{d}t + \sigma S_t v_s \,\mathrm{d}W_t.}$$

The first two state derivatives have financial names:

$$\Delta = v_s, \qquad \Gamma = v_{ss}.$$

Delta is the first-order price exposure; gamma is curvature.

Choose $\Delta_t = v_s(t, S_t)$. If changes in the stock holding are financed through the cash account, the self-financing hedged gain is

$$\mathrm{d}\Pi_t := \mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t.$$

Substitution gives

$$\boxed{\mathrm{d}\Pi_t = \left(v_t + \frac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\mathrm{d}t.}$$

Both the instantaneous Brownian shock and the unknown physical drift μ cancel.

A bookkeeping point about dynamic delta

Because Δ_t changes through time, it is generally false that

$$\mathrm{d}(V_t - \Delta_t S_t) = \mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t.$$

The right-hand side is the gain of a dynamically financed strategy. Changes in the cost of the stock position are offset by the cash account.

Example — curvature creates hedged profit and loss

For the toy claim $V_t = S_t^2$,

$$\Delta_t = 2S_t, \qquad \Gamma_t = 2.$$

The Itô formula gives

$$\mathrm{d}(S_t^2) = 2S_t \,\mathrm{d}S_t + \sigma^2 S_t^2 \,\mathrm{d}t.$$

Therefore

$$\boxed{\mathrm{d}V_t - 2S_t \,\mathrm{d}S_t = \sigma^2 S_t^2 \,\mathrm{d}t.}$$

Continuous rebalancing converts quadratic price variation into a finite-variation term.

Example — one-day gamma scale

For $\Gamma = 0.02$, $S = 100$, $\sigma = 20\%$, and $\Delta t = 1/252$,

$$\frac{1}{2}\Gamma\sigma^2 S^2 \Delta t = \frac{1}{2}(0.02)(0.20)^2(100)^2\frac{1}{252} \approx \$0.016$$

per option, before theta and financing. The number is small for one day but accumulates and is highly sensitive to gamma, volatility, and the asset level.

6.3 Preview of the Black–Scholes equation

If the delta-hedged position is locally riskless, no-arbitrage requires it to earn the bank-account rate on its value $V_t - \Delta_t S_t$:

$$\mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t = r(V_t - \Delta_t S_t)\,\mathrm{d}t.$$

Using $\Delta = v_s$ gives

$$\boxed{v_t + \frac{1}{2}\sigma^2 s^2 v_{ss} + r s v_s - r v = 0.}$$

Quadratic variation supplies the gamma term; no-arbitrage determines the growth rate of the locally riskless position. The full pricing argument is the subject of Lecture 5.

6.4 Convexity and realized variance

For f with no explicit time dependence and

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t,$$

the transformed drift is

$$f'(X_t)a_t + \frac{1}{2}b_t^2 f''(X_t).$$

Hence

shape of f	sign of f''	volatility correction
convex	$f'' \geq 0$	nonnegative
linear	$f'' = 0$	0
concave	$f'' \leq 0$	nonpositive

Volatility raises the drift of a convex transform and lowers the drift of a concave transform. The examples x^2 and $\log x$ display the two signs.

For a positive asset,

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \qquad \mathrm{d}\log S_t = \left(\mu_t - \frac{1}{2}\sigma_t^2\right)\mathrm{d}t + \sigma_t\,\mathrm{d}W_t.$$

Consequently,

$$\boxed{[\log S]_T = \int_0^T \sigma_t^2\,\mathrm{d}t.}$$

Squared high-frequency log returns estimate this integrated variance.

Example — four-day realized variance

For log returns

$$1.0\%, \quad -0.5\%, \quad 0.7\%, \quad -0.2\%,$$

the realized variance is

$$\widehat{RV} = 0.01^2 + (-0.005)^2 + 0.007^2 + (-0.002)^2 = 1.78 \times 10^{-4}.$$

Annualizing over four trading days,

$$\boxed{\widehat{\sigma}_{\text{ann}} = \sqrt{\frac{252}{4}\widehat{RV}} \approx 10.6\%.}$$

7 Guided Practice and Common Traps

7.1 Rapid practice

Exercise — three transformations

Compute:

1. $\mathrm{d}Y_t$ for $Y_t = e^{\alpha t + \beta W_t}$;
2. $\mathrm{d}Z_t$ for $Z_t = W_t B_t$ and $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$;
3. an expression without a stochastic integral for $\int_0^t e^{W_s}\,\mathrm{d}W_s$.

Short solution

$$\mathrm{d}Y_t = Y_t\left(\alpha + \frac{1}{2}\beta^2\right)\mathrm{d}t + \beta Y_t\,\mathrm{d}W_t,$$

$$\mathrm{d}Z_t = W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t + \rho\,\mathrm{d}t,$$

and

$$\boxed{\int_0^t e^{W_s}\,\mathrm{d}W_s = e^{W_t} - 1 - \frac{1}{2}\int_0^t e^{W_s}\,\mathrm{d}s.}$$

The last identity comes from integrating $\mathrm{d}(e^{W_t}) = e^{W_t}\,\mathrm{d}W_t + \frac{1}{2}e^{W_t}\,\mathrm{d}t$.

Exercise — manufacture a trigonometric martingale

Let

$$M_t = e^{at}\cos(bW_t).$$

Choose a to remove the drift, then verify that M is a true martingale on every bounded horizon.

Short solution

$$\mathrm{d}M_t = e^{at}\left(a - \frac{1}{2}b^2\right)\cos(bW_t)\,\mathrm{d}t - be^{at}\sin(bW_t)\,\mathrm{d}W_t.$$

Thus $a = b^2/2$. On $[0,T]$,

$$\mathbb{E}\int_0^T b^2 e^{b^2 t}\sin^2(bW_t)\,\mathrm{d}t \le b^2\int_0^T e^{b^2 t}\,\mathrm{d}t < \infty.$$

The stochastic integral is therefore in L^2, so M is a true martingale.

Exercise — quadratic variation after a nonlinear transform

Let $X_t = W_t^2 - t$. Find $\mathrm{d}X_t$ and $[X]_t$. Is X a true martingale on bounded horizons?

Short solution

$$\mathrm{d}X_t = 2W_t\,\mathrm{d}W_t, \qquad \boxed{[X]_t = 4\int_0^t W_s^2\,\mathrm{d}s.}$$

Since $\mathbb{E}\int_0^T 4W_s^2\,\mathrm{d}s = 2T^2 < \infty$, X is a square-integrable martingale on every bounded horizon.

7.2 Error diagnosis

Exercise — find and correct four mistakes

Suppose $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$, $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$, $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}$, and $\mathrm{d}A_t/A_t = r\,\mathrm{d}t$. Correct:

$$\begin{aligned}
\mathrm{d}f(t,X_t) &= f_x\,\mathrm{d}X_t + \frac{1}{2}b_t f_{xx}\,\mathrm{d}t,\\
\mathrm{d}(W_tB_t) &= W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t,\\
\text{zero drift} &\Longrightarrow \text{true martingale},\\
\frac{\mathrm{d}(S_t/A_t)}{S_t/A_t} &= \sigma\,\mathrm{d}W_t^{\mathbb{P}}.
\end{aligned}$$

Short solution

$$\begin{aligned}
\mathrm{d}f(t,X_t) &= \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)\mathrm{d}t + b_t f_x\,\mathrm{d}W_t,\\
\mathrm{d}(W_tB_t) &= W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t + \rho\,\mathrm{d}t,\\
\text{zero drift} &\Longrightarrow \text{local martingale first},\\
\frac{\mathrm{d}(S_t/A_t)}{S_t/A_t} &= (\mu - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.
\end{aligned}$$

The missing ingredients were, respectively: the time derivative and squared diffusion coefficient; covariation; an integrability check; and the physical excess-return drift.

7.3 A compact end-of-lecture checklist

Eight checks for every Itô calculation

1. Identify the time and state variables.
2. Compute f_t, f_x, f_{xx} before substituting the process.
3. Write every driving SDE and covariation.
4. Use $(\mathrm{d}W)^2 = \mathrm{d}t$ and $\mathrm{d}W\,\mathrm{d}B = \rho\,\mathrm{d}t$.
5. Retain every term of order $\mathrm{d}t$.
6. Group the final answer into drift and diffusion.
7. Check the domain, regularity, and integrability assumptions.
8. In a financial argument, state the probability measure and explain the financing convention.

When the classical formula is not enough

The formula in this lecture assumes $f \in C^{1,2}$. It does not apply directly at the kink of

$$f(x) = |x| \qquad \text{or} \qquad f(s) = (s-K)^+.$$

For nonsmooth convex functions, a local-time term appears in the generalized Itô–Tanaka formula. Before differentiating, locate singularities and ask whether the process can reach them.

Final synthesis

- Brownian quadratic variation makes stochastic calculus second order.
- Covariation turns correlation into a $\mathrm{d}t$-term.
- The Itô formula is a chain rule plus a curvature correction.
- Logarithms and integrating factors solve GBM, OU, and moment problems for CIR.
- Products, ratios, discounting, realized variance, and delta hedging all use the same multiplication table.
- Zero drift implies a local martingale first; suitable integrability upgrades it to a true martingale.